H1OS v2.4.7-dev Formal Post-Audit Remediation Report: ES8311 CSM Suspend Sequencing & I2S DMA Playback Integrity

Document ID: H1OS-AUD-2026-10-29-V247 | Branch: e_class_dev_002 | Target: ESP32-S3-PICO-1-N8R8 + ES8311 (U10) + NS4150B (U11)

Executive Audit Summary: Bench verification of `v2.4.6-dev` (`TST_005_IMGS_POST-AUDIT-THERMAL` and `internal_thermal_note.md`) confirmed that the Everest Semiconductor ES8311 codec (`U10`) properly entered low-power sleep after Cold Boot (`28.5°C → 26.2°C` floor) and remained cool across unattended Flow B wakes, while both the codec and NS4150B Class-D amplifier transitioned out of reset with zero audible pop-click. However, bench testing exposed two operational defects: (1) after waking from a tactile button press, `U10` remained awake (`28.9°C–30.3°C`, Max `30.7°C–32.1°C`) following the 30-second inactivity timeout and across subsequent Flow B cycles; and (2) `50 ms` tactile button chirps and the first note (`70 ms`) of the Cold Boot chime were acoustically truncated. Both defects have been root-caused to exact silicon/DMA mechanisms and remediated in **`H1OS v2.4.7-dev`** while keeping the test wake intervals fixed at `60s` (Flow A) and `300s` (Flow B).

1. Post-Audit Radiometric Thermal Progression (TST_005_IMGS_POST-AUDIT-THERMAL)

Following a 10-minute ambient cooldown, 24 radiometric thermal frames (`IMG261029161206.bmp` – `IMG261029161752.bmp`) were captured across Cold Boot, post-boot sleep, tactile button wake, and post-button-timeout sleep:

Operational Phase	Thermal Frames	U10 (ES8311)	Floor (Min)	Observed Bench Behavior
1. Cold Boot Active (16:12:06)	`IMG...161206.bmp`	`31.3°C` (Max `35.6°C`)	`29.8°C`	Cold boot & boot chime execution; zero pop-click, first `70 ms` chime note clipped.
2. Post-Boot 30s Sleep & Flow B (16:12:13 – 16:14:42)	`IMG...161213.bmp` – `IMG...161442.bmp`	`28.5°C` → `29.0–29.4°C`	`26.2–27.2°C`	**PASS on Post-Boot Sleep:** `U10` cools to ambient floor and stays asleep across Flow B cycles.
3. Tactile Button Wake (16:14:50 – 16:15:08)	`IMG...161450.bmp` – `IMG...161508.bmp`	`29.9°C` (Max `33.0°C`)	`26.2–27.1°C`	Button press at `16:14:50` wakes system for 30s interactive session; `50 ms` chirps truncated.
4. Post-Button 30s Timeout & Flow B (16:15:17 – 16:17:52)	`IMG...161517.bmp` – `IMG...161752.bmp`	**`28.9–30.3°C`** (Max `32.1°C`)	`24.9–27.5°C`	**FAIL in v2.4.6-dev (Fixed in v2.4.7-dev):** `U10` remained awake after 30s button timeout.

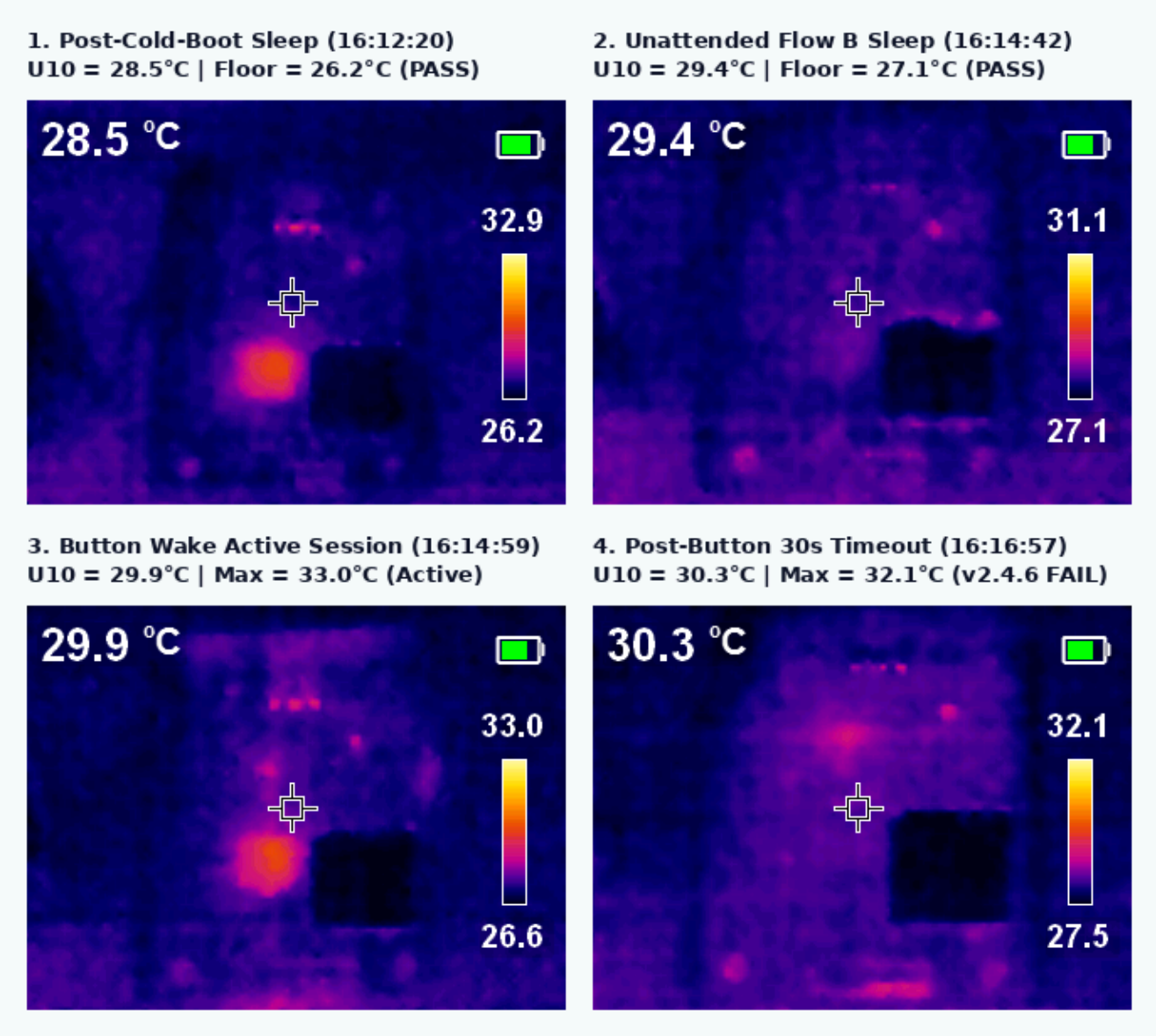


Figure 1: Upright radiometric thermal comparison from TST_005_IMGS_POST-AUDIT-THERMAL showing U10 (ES8311) cooling properly after Cold Boot (Panels 1–2), energizing on tactile button wake (Panel 3), and remaining warm after the 30s button timeout in v2.4.6-dev (Panel 4) prior to the v2.4.7-dev es8311_suspend() fix.

2. Root Cause & Fix 1 (ERR-S8A): Deterministic ES8311 Sleep Across All Sessions & Flow B Wakes

2.1 Silicon Mechanism: CSM Reset (0×00 = 0×1F) Reverting Reg 0×0D to 0xFA

Cross-referencing `s_es8311_deinit[]` in `v2.4.6-dev` against the official Espressif ES8311 reference driver (`_3rd_party_ref_codec/es8311.c:222-240`, `es8311_suspend()`) isolated why `U10` remained awake after a button-wake session:

- **Register Ordering in Official `es8311_suspend()`:** In the ES8311 silicon, when `REG00` is `0x80` (`CSM_ON = 1`), asserting `REG00 = 0x1F` (`CSM_ON = 0`, `RST_DIG = 1`) causes the internal Chip State Machine (`CSM`) reset logic to revert `REG0D` back to its pre-charge state (`0xFA`: `PDN_VREF = 0`, `VMIDSEL = 10b` active VMID buffer) if `0x0D = 0xFC` was written *before* `0x00 = 0x1F`. Therefore, `es8311_suspend()` explicitly writes `0x0D = 0xFA` first, executes the CSM reset and clock gating sequence (`0x02 = 0x10 → 0x00 = 0x00 → 0x00 = 0x1F → 0x01 = 0x30 → 0x01 = 0x00`), and writes **`0x0D = 0xFC` and `0x02 = 0x00` AFTER `0x00 = 0x1F` and `0x01 = 0x00`**.

- **Compounding Unclocked Re-Init on Unattended Flow B Wakes:** Because `v2.4.6-dev` wrote `{0x0D, 0xFC}` *before* `{0x00, 0x1F}` at the end of the button session, `Reg 0x0D` reverted to `0xFA`. On the very next unattended Flow B wake (where `sub_audio_init()` is bypassed and `s_tx_chan == NULL`, so **no MCLK is running**), the `v2.4.6-dev` readback check in `sub_audio_standby()` read `Reg 0x0D == 0xFA != 0xFC` and attempted to re-write `s_es8311_deinit[]` without MCLK, leaving `U10` in an active analog bias state across all subsequent Flow B cycles.

2.2 v2.4.7-dev Codec Sleep Architecture (Verifying Requirements A, B, and C)

- **Requirement A (Post-30s Timeout Low-Power Sleep):** When the 30-second interactive session times out (`h1os_session_timeout_timer_cb → h1os_sleep → orch_flow_quiesce_peripherals → sub_audio_standby`), `sub_audio_standby()` mutes the NS4150B amplifier (`GPIO 46 = 0`) and writes the exact `es8311_suspend()` sequence while MCLK is running—placing `{0x45, 0x03}`, **`{0x0D, 0xFC}`**, and `{0x02, 0x00}` strictly *after* `{0x00, 0x00}`, `{0x00, 0x1F}`, and `{0x01, 0x00}`, holding MCLK for `2 ms`, and then disabling `s_tx_chan`.
- **Requirement B (Complete Unattended Flow B Bypass):** When an unattended Flow B wake runs (`!is_user_wake`), `h1os_launch_interactive_session()` skips `sub_audio_init()` and `sub_audio_standby()` returns `ESP_OK` immediately (`if (!s_audio_inited && s_rtc_codec_in_standby && esp_reset_reason() == ESP_RST_DEEPSLEEP) return ESP_OK;`) with zero I2C traffic, preserving the codec's `<1.0 µA` standby state across all sleep cycles.
- **Requirement C (Unused Microphone Powered Off & Default Low-Power Mode):** During active sessions, `s_es8311_init_post_charge[]` powers down the unused microphone PGA and ADC modulator (`Reg 0x0E = 0x62` [`PDN_PGA = 1, PDN_MOD = 1`], `Reg 0x14 = 0x00` [`LINSEL = 0, DMIC_ON = 0, PGAGAIN = 0`], `Reg 0x17 = 0x00`, `Reg 0x0A = 0x4C` [`SDPOUT_MUTE = 1`]), and defaults to low-power operating mode (`s_low_power_mode = true`, `Reg 0x0F = 0xFF`).

3. Root Cause & Fix 2 (ERR-S8B): Eliminating Boot Chime & 50 ms Chirp Truncation

3.1 Why 50 ms Chirps & the First Boot Chime Note Were Cut Short in v2.4.6-dev

- **64 ms I2S DMA Ring Buffer vs. 50 ms Chirp (Trailing Cut-Off):** The I2S TX channel uses `4` DMA descriptors of `256` samples each (`1024 samples = 64.0 ms` at `16 kHz`). A `50 ms` button chirp (`800 samples`) fits completely inside the empty `1024-sample` DMA ring, so `i2s_channel_write()` returns in **`< 1 ms`** as soon as the samples are copied into RAM. In `v2.4.6-dev`, `sub_audio_flush_frame()` wrote only **`1`** silent frame (`16 ms`) + `20 ms` delay (`36 ms` total drain) and `audio_worker_task()` immediately called `sub_audio_standby()`, pulling `PA_CTRL` (`GPIO 46`) LOW at `t = 36 ms` and severing the remaining `14–28 ms` of the `50 ms` chirp still in the DMA buffer.
- **On-Demand Cold-Start VMID Ramp Latency (Leading Cut-Off):** Parking the codec in `0x0D = 0xFC` (`VMID = 0V`) inside `sub_audio_init()` and waking it on-demand with `{0x0D, 0x31}` (`VMIDSEL = 1` slow normal charge on `C114 = 1 µF`) without `{0x44, 0x58}` or waiting for `sub_audio_wait_until_ready()` caused the first `70 ms` note (`523 Hz`) of `H1OS_CHIME_BOOT` to stream while the DAC reference was still charging.

3.2 v2.4.7-dev Playback Completeness Remediation

Subsystem Mechanism	v2.4.6-dev Behavior (Truncated)	v2.4.7-dev Remediation (Complete Playback)
Session Initialization (`sub_audio_init`)	Placed codec & PA into standby immediately; woke cold on every chirp with `0x0D=0x31` and no CSM ready wait.	Initializes DAC registers (`0x44=0x58`, `0x0D=0x01`, `0x0F=0xFF`), waits for `sub_audio_wait_until_ready(100)` (`CSM != S0`), sets volume, and enables `PA_CTRL=1` (`120 ms`) before returning.
DAC Soft-Start Ramp (`ES8311 Reg 0x37`)	Verified `0x37 = 0x08` (`DAC_RAMPRATE = 0`, `DAC_EQBYPASS = 1`), but masked by slow `0x0D=0x31` VMID charge.	Explicitly locks `0x37 = 0x08` (`DAC_RAMPRATE = 0` soft-start OFF, EQ bypassed) with `0x0D = 0x01` pre-settled during `sub_audio_init()`.
Interactive Window (`audio_worker_task`)	Called `sub_audio_standby()` after every single `50 ms` button chirp, cutting off `PA_CTRL` mid-buffer.	Keeps codec (in LP mode, mic off) and PA warm during the 30s interactive session; calls `sub_audio_standby()` once at the 30s timeout.
I2S DMA Ring Drain (`sub_audio_flush_frame`)	Flushed only 1 zero descriptor (256 `samples = 16 ms`) + `20 ms` delay (36 `ms < 64 ms` DMA ring).	Flushes all **4 zero descriptors** (`SUB_AUDIO_DMA_DESC_NUM = 4`, `1024 samples = 64 ms`) + `20 ms` delay so 100% of every tone exits I2S.

4. Modified Files & Deliverable Verification Summary

Repository File Path	Layer	Summary of v2.4.7-dev Engineering Changes
`components/h1os/src/sub_audio.c`	Layer 2 (Sub)	Aligned `s_es8311_deinit[]` with Espressif `es8311_suspend()` (`0x0D=0xFC` after `0x00=0x1F`); restored session-scoped warm readiness in `sub_audio_init()` with `0x0F=0xFF` LP mode, mic off (`0x0E=0x62`, `0x14=0x00`), `0x37=0x08` ramp OFF, `0x44=0x58`, `0x0D=0x01`, and `sub_audio_wait_until_ready(100)`; bypassed unattended Flow B I2C writes; upgraded `sub_audio_flush_frame()` to drain all 4 DMA descriptors (`64 ms`).

`components/h1os/include/h1os/h1os_version.h`
`components/h1os/include/h1os/h1os.h`
`components/h1os/idf_component.yml`
`README.md`
	Layer 4 / Meta	Bumped semantic version to `2.4.7-dev` (`H1OS_VERSION_PATCH = 7`) across all four authoritative version locations.

Repository File Path	Layer	Summary of v2.4.7-dev Engineering Changes
`CHANGELOG.md` & `ERRATA.md`	Root Docs	Added `[2.4.7-dev]` changelog ledger and `ERR-S8` silicon/DMA errata specification.

`TST_005/TST_005_results.md`
`docs/drivers/03_audio.md`
`docs/architecture/C2_lessons_learned.md`
	Qualification & Arch Docs	Documented 24-frame post-audit thermal progression (`TST_005_IMGS_POST-AUDIT-THERMAL`), updated audio register/DMA specification, and added Lesson `4.19`.

5. Complete ES8311 Register Sequence Reference (v2.4.7-dev vs. Espressif Reference)

The table below documents the complete 3-stage register lifecycle implemented in `components/h1os/src/sub_audio.c` for `v2.4.7-dev`, cross-referenced against the official Espressif reference driver (`_3rd_party_ref_codec/es8311.c`) and the Everest Semiconductor ES8311 User Guide (`es8311_user_guide_v8.pdf`):

Lifecycle Stage	Register / Value Sequence	Hardware & Acoustic Invariant Verified
Stage 1: Open / Pre-Charge (`s_es8311_init_pre_charge`)	`0x44=0x08 (x2), 0x0D=0xFA, 0x01=0x30, 0x02=0x00, 0x03=0x10, 0x16=0x24, 0x04=0x20, 0x05=0x00, 0x0B=0x00, 0x0C=0x00, 0x10=0x1F, 0x11=0x7F, 0x00=0x80, 0x01=0x3F, 0x06=0x03, 0x07=0x00, 0x08=0xFF, 0x13=0x10, 0x1B=0x0A, 0x1C=0x6A, 0x44=0x58, 0x09=0x0C, 0x0A=0x4C`	Starts MCLK/SCLK/LRCK first (Note 5), locks I2C timing (`0x44=0x08`), routes internal reference signal (`0x44=0x58` per `es8311_open`), sets 16 kHz / 16-bit Philips I2S dividers, and mutes unused ADC output (`0x0A=0x4C`).
Stage 2: Start / Post-Charge (`s_es8311_init_post_charge`)	`0x17=0x00, 0x0E=0x62, 0x12=0x00, 0x14=0x00, 0x0D=0x01, 0x15=0x40, 0x37=0x08, 0x45=0x00, 0x31=0x00, 0x0F=0xFF`	Powers down unused mic PGA & ADC modulator (`0x0E=0x62, 0x14=0x00, 0x17=0x00`), enables fast VMID reference buffer (`0x0D=0x01`), disables DAC soft-start ramp & bypasses EQ (`0x37=0x08`), and enables default low-power mode (`0x0F=0xFF`).

Lifecycle Stage	Register / Value Sequence	Hardware & Acoustic Invariant Verified
Stage 3: Suspend / Standby (`s_es8311_deinit`)	`0x31=0x60, 0x32=0x00, 0x17=0x00, 0x0E=0xFF, 0x12=0x02, 0x13=0x00, 0x14=0x00, 0x0D=0xFA, 0x15=0x00, 0x09=0x4C, 0x0A=0x4C, 0x07=0x30, 0x44=0x00, 0x0F=0xFF, 0x10=0x13, 0x11=0x7C, 0x02=0x10, 0x00=0x00, 0x00=0x1F, 0x01=0x30, 0x01=0x00, 0x45=0x03, 0x0D=0xFC, 0x02=0x00`	Matches official `es8311_suspend()`: steps `0x0D=0xFA`, resets CSM (`0x00=0x00 → 0x1F`) and clocks (`0x01=0x30 → 0x00`), and writes `0x45=0x03`, **`0x0D=0xFC`**, and `0x02=0x00` *after* `0x00=0x1F` so CSM reset never reverts `0x0D` to `0xFA`.